

PRITHVI KRISHNASWAMY

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Engineering Skills: Altium Designer, LTSpice, Python, Cadence Virtuoso/Spectre, MATLAB, Simulink, PLECS, Ansys HFSS

Education Credentials

Bachelor of Science: Electrical Engineering | **University of Washington (UW)**, Seattle, WA | Graduation: 06/2025
Minor: Applied Mathematics | **Coursework: Circuit Theory, Energy Systems, Applied Electromagnetics** | **GPA: 3.57**
Master of Science: Electrical Engineering | **University of Washington (UW)**, Seattle, WA | Graduation: 06/2027
Coursework: Power Electronics, Control Systems, Analog Integrated Circuits, Antenna Design | **GPA: 3.82**

Professional Experience

Avionics Hardware Design Engineering Intern (Incoming) | **Stoke Space** | Kent, WA | Sep 2026 – Dec 2026

- Design & Testing **Servo Controllers** For [Nova](#): Stoke Space's fully reusable launch vehicle!

EV Electrical Design Engineering Intern | **Rivian/Volkswagen Technologies** | Irvine, CA | Jun 2026 – Sep 2026

- Tuned reliability of **Pierce Oscillator Circuit** across 10 Zonal Controllers, 4 vehicle families & up to ~100,000 vehicles.
- Characterized **12V/48V HSDs** & validated [RCP](#) Emulation for Zonal-Controlled edge nodes, advancing a SW-less architecture eliminating local MCUs, device-resident application software/OTA burden & reduced wiring complexity
- Normal-Load Efficiency/Thermal Performance characterization of a **2-Phase 48V-to-15V GaN DC-DC Converter** across 1-34 A loads, 36V/48V/54V inputs, and -40 C to 85 C test corners. Demonstrated 98.03% efficiency
- Tested & Updated a **48V-to-12V short-circuit protection test board** for 12V HSDs against 440 A transients under a 48 V Bus Fault. Compared component-level surge-energy, clamping voltage and recovery performance

Aircraft Electrical Design Engineering Intern | **Shield AI** | Dallas, TX | Jan 2026 – Jun 2026

- Hardware Bring-up, DVT & Qualification of **V-BAT [ViDAR](#) eXtensible Processing Module (XPM)**. Validated Electrical, Thermal, OS & Mission Software Parameters for 1st stage Experimental Flight Test (XFT)
- Prototyped DAQ Interposer PCB to validate connector stability. Scripted Python vibration tests & logged data using LabVIEW
- Owned EE responsibilities for **V-BAT Battery Management System (BMS)** enabling cell-level Temperature, Voltage, State of Health (SOH), and State of Charge (SOC) monitoring. Added enhanced Protection and Sequence Mechanisms. Qualified Thermal, Vibration & Quiescent Current Discharge Performance to match design/operating parameters

Datacenter Hardware Design Engineering Intern | **Nvidia** | Durham, NC | Jun 2024 – Sep 2024

- Designed a simplified GPU-emulator PCB in **Cadence Allegro** that emulated key power, signal, and I²C interfaces of [the GB200 server switch tray](#), cutting test-fixture expenses by ~85% and accelerating bring-up cycles.
- Qualified **Single/Multi Phase Buck & LDO rails** on [50xx GPU](#) SLT boards. Characterized FRA Bode Plot, Transient, Efficiency, Input Droop, Output Ripple, and AMR measurements to achieve ≥55° phase margin and <30 mVpp at rated load.
- Re-architected a Power Adapter PCB for 100+ engineers— transitioned to CEM5 high-current connectors, added OVP/UVF, inrush control - to improve safe & flexible AC/DC and DC/DC power-conversion for DGX/MGX server platforms.

Plasma Etch Silicon Engineering Intern | [Intel](#) | Hillsboro, OR | Jan 2023 – Dec 2023

- Innovated a custom **Deep Reactive Ion Etching (DRIE) [Bosch Process](#)**, achieving 50:1 to 100:1 aspect ratio trenches and holes for MEMs Through-Silicon Vias (TSVs). Enhanced etching precision by 3x baseline on 20 validated silicon wafers.

Automotive Technician & App Developer | **Singapore Armed Forces** | Singapore | Oct 2019 – July 2021

- Mechanical & Electrical maintenance on 4 different fleets of 20 vehicles. Conducted **Failure Analysis**, debugging & validation of Batteries, Filters & Drivetrains. Catalogued defects & ordered necessary replacements.
- Constructed a user-friendly front-end interface for a multiplatform application aimed at increasing workshop safety awareness; designed back-end connections that ensured secure handling of user input data for over 100 users
- Beta-tested the application with documentations and [training videos](#) that increased user adoption by a factor of 4

Projects

Electronics Engineer | [UW Formula Motorsports \(FSAE\)](#) | Seattle, WA | Jul 2025 – Jun 2026

- Architecting an [Integrated Powertrain Module](#) to reduce HV harness length, simplify packaging, and improve EMI behavior
- Spearheading architecture & design of a custom 900W, isolated [Resonant LLC DC – DC Converter](#) reducing battery costs by 90%, while achieving dual compatibility with 400V & 600V HV pack architectures
- Renovating Schematic & Layout, in Altium, of a custom 26 kVA, 540V DC high-performance [Quad – Motor Inverter](#) to decrease car weight by 11kg and increase efficiency up to ~98 %

Extracurriculars

Teaching Assistant | **UW Department of Electrical & Computer Engineering** | Seattle, WA | Jun 2023 – Jun 2025

- Built a [line-following robot platform](#) used by 100+ students, to teach full-stack prototyping (PCB → firmware → validation).
- Coordinated 100+ office-hour sessions (~30 students/week) and streamlined grading for labs & HW

Lead Research Assistant | [Washington Nanofabrication Facility](#) | Seattle, WA | May 2022 - Current

- Validated Vision **Reactive Ion Etcher (RIE)** quality & reliability. Subjected ~100 silicon wafers across various processes, verifying trench depth & uniformity using a **Scanning Electron Microscope (SEM)**. Published process notes for lab users.
- Doubled etching rate of Oxford **Inductively Coupled Plasma (ICP) Etcher** for PECVD & Thermal Oxide deposited Silicon Dioxide wafers. Documented parameters for reproducibility.